

11/07/2026

COMPUTER VISION

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SHORT ANSWERS.

1. * Computer Vision uses algorithms and AI to analysis Images and videos, allowing computers to identify objects, Patterns and Scenes like human vision.

a) * Extract meaningful informations from images.

* Recognize and understand objects and Scenes.

b) * Face Recognition.

* Medical image analysis.

2. * Computer vision provides image enhancement, feature extraction, Object detection, Recognition and Scenes understanding.

a) * Feature Extraction is the process of identifying important enhancement characteristics such as edges, corners, features or shape of the images.

b) * It helps identify and classifies Objects accurately, enabling intelligent decision-making.

3. * It involves basic operations such as noise removal and image enhancement and filtering.

b) MID LEVEL: Image Segmentation and Feature extraction.

HIGH LEVEL: Object Recognition and Scenes understanding.

4. * A digital image is represented as an grid of pixels, which each pixels store identity or color information.

a) * A pixel is a smallest element of a digital image that represents a single color or intensity value.

b) * High resolution provides more details and better image quality.

5. * Image Sensing converts real-world scene into a digital image from using sensors and devices.

a) IMAGE SENSOR:

* It captures light and converts it into Electrical Signals.

* The two Image acquisition devices are :

- Digital camera
- Scanner.

6. SAMPLING:

a) * Sampling converts a continuous image into discrete pixels is called Sampling.

b) * The purpose of quantization in image processing is to assigns the intensity values to pixels, defining image quality.

7. * Image formation describes how light interacts with objects and is captured by a camera.

a) PERSPECTIVE PROJECTION:

* It represents a 3D scene on a 2D image plane based on viewpoint.

b) ROLE OF ILLUMINATION:

* Lighting affects brightness, contrast and visibility of objects.

8. IMAGE PROCESSING:

- a) * It involves enhancing and modifying images without understanding them.

IMAGE PROCESSING:

- b) * Image processing focuses on enhancement.
- * Image processing outputs images.

COMPUTER VISION:

- * It focuses on interpretation.
- * Computer vision outputs decisions.

9. HEALTH CARE:

- a) * Used in medical imaging (X-rays) (MRI) to detect diseases.

b) AUTONOMOUS VEHICLES:

- * used for object detection, lane tracking, and obstacle avoidance.

10. PIXELS AS BUILDING BLOCKS:

- a) * Pixels store intensity / color values and form the entire image quality and analyse image structure.

SAMPLING & RESOLUTION:

* Higher Sampling and resolution improve image quality and analysis accuracy.

